

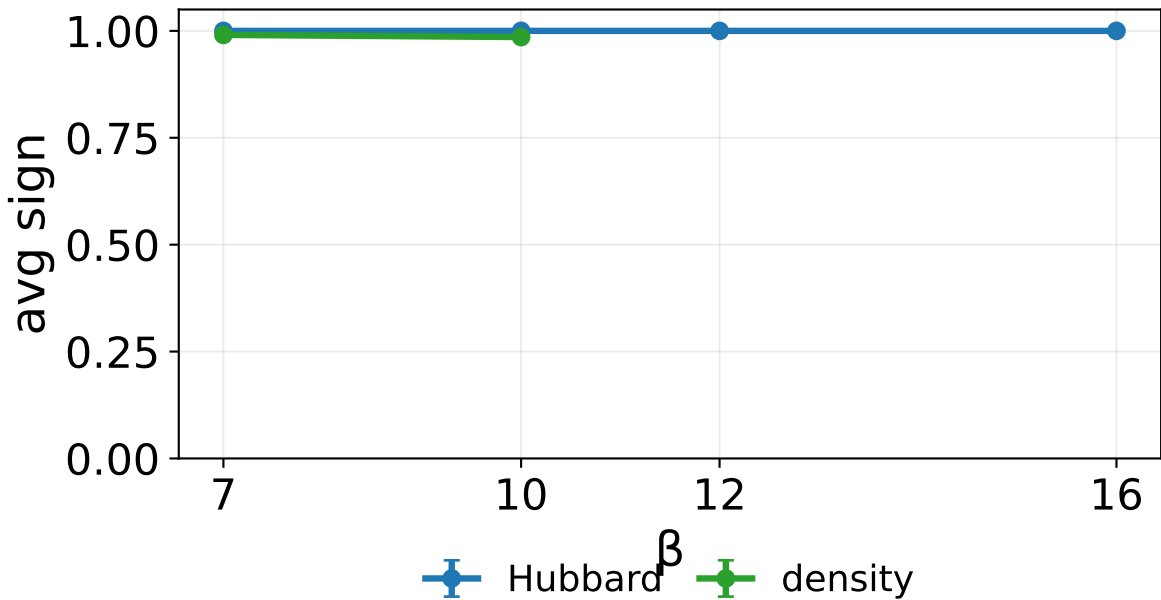
Pi-flux QSH Hubbard-Kanamori DQMC: average sign summary

Lx=8, Ly=4, periodic clean system; U=1.0, J_H=0.25, t=1.0, λ=0.2, dtau=0.1; Ntherm=2000, Nmeas=10000 per rank, 32 ranks.

Current summary generated 2026-06-10 15:05. μ=0.5 for Hubbard; μ=0.875 for Kanamori tiers. No μ scan.

At physical half filling

boundary condition	β	Hubbard avg sign	density avg sign	spin-flip avg sign	full avg sign
periodic	7	1.000	0.991	pending	pending
periodic	10	1.000	0.986	pending	pending
periodic	12	1.000	pending	pending	pending
periodic	16	1.000	pending	pending	pending



Takeaway

- Hubbard-only remains sign-free at half filling.
- Density Kanamori stays high in completed rows.
- Completed summary rows: 6/16.